Pithani Avinash Kumar

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EDUCATION

- **Indian Institute of Technology Bombay** | IIT Bombay *[2025–2027]*
  M.Tech in Geoinformatics, CSRE dept. **CPI: 8.12/10**
- **Vignan's Institute of Information Technology (VIIT)** | JNTU-GV *[2021–2025]*
  B.Tech in Computer Science and Engineering, CSE dept. **CGPA: 8.83/10**

EXPERIENCE

- **Web Development (Full-Stack) Intern** | Neuxa *[May'25–Oct'25]*
  - Developed reusable frontend components using **React Native, Expo, and JavaScript** for mobile workflows
  - Translated **UI/UX designs and wireframes** into consistent, responsive & user-friendly mobile interfaces
  - Collaborated with backend developers to integrate **REST APIs** and application data into UI components
- **Full-Stack Web Developer Intern** | Uniconn *[Sep'24–Oct'24]*
  - Developed and optimized MERN features for Uniconn Platform v2, improving API efficiency and responsiveness
  - Built modular web components using **TypeScript** and modern frontend frameworks in sprint-based workflows

KEY PROJECTS

- **SwapCharGo** | *B.Tech EPICS Project* *[Mar'23–Aug'23]*
  - Developed a **full-stack platform** for EV charging & battery swapping services with **location-based discovery**
  - Implemented **authentication, user management**, and **role-based access** for Electric Vehicle (EV) users and hosts
  - Built **station management** and **booking workflows** to support seamless charging and battery swapping services
  - Integrated **Mapbox APIs** into the application for **interactive mapping, nearest-station search** & discovery
- **Airbnb Clone** | *Self Project* *[Jun'25–Jul'25]*
  - Engineered an accommodation platform using **MVC architecture**, **server-side rendering**, and modular routing
  - Implemented **authentication, authorization, session management, CRUD, file uploads**, and **RBAC**
  - Built **reusable backend modules** & **custom middleware** for request handling, validation & API development
  - Designed and developed **RESTful APIs** enabling secure communication between frontend and backend services
- **Real-Time Decision Making in Crisis Management** | *B.Tech Main Project* *[Jan'25–Apr'25]*
  - Developed a **Flask-based web application** for flood prediction with **authentication, SQLite**, and **email alerts**
  - Built an **end-to-end ML pipeline** for **data preprocessing, feature engineering**, training, and **inference**
  - Evaluated **5 ensemble models** via **5-fold CV**; selected **CatBoost** achieving **96.26% prediction accuracy**
- **GeoPulse** | *M.Tech Course Project – GNR629* | *Prof. Surya S. Durbha* *[Mar'26–May'26]*
  - Developed client-server GIS platform using **GeoServer, PostGIS, Tomcat**, and **OpenLayers** for **OGC services**
  - Built workflows for **XML parsing, spatial data requests, multi-layer visualization** and **configurable access**
  - Extended system with **OGC SOS client** supporting **SensorML, spatial/temporal filtering**, and **mapping**
- **GIS Interface for Calibration of a Hydrological Model** | *M.Tech Thesis* | *Prof. Pritam Das* *[Jul'26–Present]*
  - Developing a **full-stack GIS app** for **calibration and monitoring** of a physically based hydrological model
  - Architecting a **modular workflow system** integrating GIS interfaces, execution management, and **HPC**
  - Designing an extensible **software architecture** for scientific computing workflows, emphasizing modularity

SCHOLASTIC ACHIEVEMENTS

- Secured **2nd Position** in **ALGO.HACKER Summer Challenge 2026**, Dept. of CSE, IIT Bombay *[Jun'26]*
- Secured **98%ile** in GATE Computer Science and Information Technology (CS) 2025 *[Mar'25]*
- Secured **92%ile** in GATE Data Science and Artificial Intelligence (DA) 2025 *[Mar'25]*

LEADERSHIP & RESPONSIBILITIES

- **Team Lead** | Vision Crafters | Smart India Hackathon (SIH 2023) *[Sept'23–Oct'23]*
  - Led a **6-member team** from ideation to prototype development under **Renewable/Sustainable Energy** theme
  - Presented the **SwapCharGo** prototype at institute-level **SIH** evaluation, securing nomination to represent VIIT
- **Teaching Assistant** | *GNR653: Data Analysis Methods for Geospatial Applications* | *IIT Bombay* *[Jul'26–Present]*
  - Assist in tutorials and hands-on sessions on **open-source scientific computing** and **geospatial data analysis**
  - Support in practical exercises involving **version control**, resolve queries, and facilitate collaborative workflows

COMPETITIVE PROGRAMMING PROFILES

- Participated in **100+ contests** and solved **500+ problems** collectively on platforms like **LeetCode** & **CodeChef**

TECHNICAL SKILLS

Programming Languages	: C++, C, Python, SQL, JavaScript, Java
Core Computer Science	: Data Structures, Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Backend & Databases	: Node.js, Express, RESTful APIs, MongoDB, PostgreSQL/PostGIS, SQLite, FastAPI
Frontend & Mobile	: HTML, CSS, TypeScript, React, Tailwind CSS, Material UI, React Native, Expo, EJS
Data & AI	: MySQL, Power BI, Excel, Pandas, NumPy, LLM APIs, Prompt Engineering
Tools & Platforms	: Git, Linux (Fedora), Docker, CI/CD, AWS, GeoServer, OpenLayers